

FIST Meets the Semantic Knowledge Graph: Aligning Fast, Inexpensive, Simple, Tiny with Dinis Cruz's G³ Approach

by Dinis Cruz and ChatGPT Deep Research, June 2025

Introduction

Modern engineering projects thrive on agility, thrift, and simplicity. These values are encapsulated in the FIST framework – **Fast, Inexpensive, Simple, Tiny** – originally coined in the defense acquisition context to promote rapid, low-cost innovation ¹ ². FIST is described as a decision-making framework that guides practitioners toward streamlined processes, accelerated timelines, and restrained scope ³. Meanwhile, contemporary knowledge-engineering methodologies, such as Dinis Cruz's work with **Semantic Knowledge Graphs** and the **G³ (Graphs of Graphs of Graphs)** concept, emphasize flexible, dynamic structures for organizing information. This white paper provides a comparative analysis of how the FIST principles align with and are exemplified by Dinis Cruz's Semantic Knowledge Graph approach. We explore conceptual mappings between the two, illustrate the comparison with concrete examples, and extract actionable takeaways. The goal is to show that although FIST emerged in a military-tech context, its core ideas resonate strongly with modern knowledge graph engineering practices in theory and practice.

Overview of FIST Principles (Fast, Inexpensive, Simple, Tiny)

FIST is a set of values and design principles aimed at delivering capabilities **quickly, cheaply**, and with minimal complexity or bloat ⁴ ⁵. In practical terms, a FIST project is one that deliberately keeps schedules short (Fast), budgets small (Inexpensive), designs straightforward (Simple), and scope or team size limited (Tiny). These elements reinforce each other – *if a project is fast, it inherently has a tiny schedule; if inexpensive, a tiny budget; if simple, a tiny degree of complexity* ⁶ ⁷. Crucially, FIST is about **restraint**: resisting the temptation to add time, money, or features as a default solution ⁸. Instead, it advocates fixed or tight schedules with flexible requirements (deliver the most important features first) ⁵.

FIST also challenges the notion that bigger and more complex is better. Conventional wisdom might equate complexity with sophistication, but FIST posits that unnecessary complexity is a sign of immature design, not something to brag about ⁹. True sophistication is found in elegant simplicity ¹⁰. By keeping systems and processes simple and small, FIST projects aim to minimize overhead and risk. As Lt. Col. Dan Ward (FIST's originator) summarized, *"adding people to a late project makes it later,"* and extending schedules or scope often has negative impacts ⁸. Thus, FIST enforces discipline: **short deadlines, small budgets, minimal requirements**, and an emphasis on doing the common things uncommonly well.

Over the years, the FIST philosophy has proven effective in domains ranging from software to hardware. It has a strong overlap with Agile methods – in fact, Agile practices like Scrum or Extreme Programming that **reduce cost, duration, and complexity** are considered part of the FIST

“toolbox” ¹¹ ¹². In summary, FIST establishes a culture of speed, thrift, simplicity, and restraint, steering projects away from sprawling, slow, expensive efforts toward leaner, faster, and more focused endeavors ⁷ ¹³.

Dinis Cruz’s Semantic Knowledge Graph and G³ Methodology

Dinis Cruz’s research and engineering methodologies center on leveraging **Semantic Knowledge Graphs** – rich, flexible networks of information – to capture and organize knowledge dynamically. A Semantic Knowledge Graph (SKG) is essentially a knowledge graph imbued with explicit meaning via an ontology or schema. Unlike rigid databases, an SKG’s schema is designed to **evolve and change over time** without massive rework ¹⁴. This flexibility is crucial: as Dinis notes, graphs work and scale best when they have a schema that can adapt, something traditional relational databases struggle with ¹⁴. The SKG approach treats knowledge as living data: nodes represent concepts/entities and edges represent semantic relationships, all of which can be refactored, extended, and even *diffed* or version-controlled like software ¹⁴. In practice, this means the knowledge graph can continuously improve – much as code is iteratively improved – ensuring the system remains simple and relevant rather than ossifying under outdated complexity.

A key part of Cruz’s methodology is the concept of **G³, or Graphs of Graphs of Graphs**. G³ is an approach to manage multiple interconnected graphs across different domains or perspectives ¹⁵. Instead of trying to force everything into one “uber” master ontology, G³ embraces a federation of graphs – essentially *ontologies of ontologies* and *taxonomies of taxonomies* – that communicate with each other ¹⁵. In other words, each domain or stakeholder can have its own tiny knowledge graph (with its own schema suited to its context), and these graphs can be linked or translated via higher-level graphs. This approach acknowledges that there is no single one-size-fits-all model of knowledge; rather, value comes from connecting multiple small graphs and viewpoints. *“There isn’t a master Ontology or Taxonomy,”* Cruz explains, *“what we have are ways to communicate and connect multiple graphs, domains, and ultimately different points of view.”* ¹⁵. This G³ philosophy exemplifies **modularity and restraint**, avoiding a monolithic design in favor of many small, interoperable pieces – very much in the spirit of “Tiny” and “Simple” principles.

In implementing Semantic Knowledge Graphs, Dinis Cruz’s engineering practices also mirror FIST-like values. His projects use **open-source tools** and lightweight, cloud-native architectures to stay *inexpensive* and *fast*. For example, Cruz co-developed a stack using **OSBot** (for orchestrating interactive AI dialogues) and **MGraph-DB** (a lightweight JSON-based graph database), deployed on a serverless infrastructure ¹⁶. This means knowledge graphs are built on-demand in memory and persisted to simple cloud storage, allowing the system to scale out as needed without costly dedicated servers ¹⁷ ¹⁸. The entire pipeline is designed to be elastic and efficient: when a user initiates a session or query, serverless functions spin up to build or query the semantic graph, then spin down – consuming resources only when needed ¹⁷. By using standard and open technologies, the solution avoids proprietary vendor lock-in and remains easily extensible ¹⁹ ²⁰. In fact, Cruz has successfully used this stack in production projects, noting how it enabled maintaining *“multiple interconnected Semantic Knowledge Graphs that serve different stakeholders... without forcing anyone to learn a standardized language,”* with the GenAI (Generative AI) acting as *“the translator, the connector, and the catalyst for continuous improvement.”* ²⁰. This setup empowers flexible knowledge sharing: each user or team gets a tailored graph (their own small piece of the puzzle), and AI bridges these pieces when needed – all running efficiently at scale ²¹.

Another hallmark of Cruz’s methodology is **interactive, AI-driven development of the knowledge graph**. Instead of designing a giant static questionnaire or data model upfront, Cruz’s Semantic

Knowledge Graph workflow often uses large language models (LLMs) to conduct adaptive Q&A sessions with users, building the graph in real time. At the start, the graph might have only a single node (representing, say, a user or a topic), and the LLM begins asking the user context-relevant questions ²² ²³. Each answer is parsed and added as new nodes/edges in the graph, and subsequent questions adapt based on the growing graph structure ²² ²⁴. This **human-in-the-loop** process continues until a rich, personalized graph of knowledge is constructed. Throughout, the user validates and corrects the information (ensuring simplicity and accuracy), effectively co-creating the knowledge base. All data ends up structured as a semantic graph, built dynamically using AI, and *“every conversation [becomes] a living, evolving graph.”* ¹⁶ ²⁵. The result is a system that captures knowledge much faster and in a more engaging way than traditional manual form-filling, and one that remains *tiny* in scope at each step (asking only relevant questions, one at a time) while growing organically.

In summary, Dinis Cruz’s Semantic Knowledge Graph + G³ approach emphasizes **fast iteration, cost-effective open tools, simplified knowledge modeling, and modular “tiny” knowledge units**. Next, we delve into a direct comparison to see how each FIST principle is reflected in this approach.

Mapping FIST Principles to Cruz’s Knowledge Graph Approach

Speed and Agility (“Fast”)

FIST places a premium on speed: short development cycles and delivering useful capability *“on operationally relevant timelines”* ²⁶. In practice, FIST projects keep schedules aggressively tight, demonstrating that rapid delivery is not only possible but often more successful than protracted efforts. For instance, the Air Force’s Project Liberty surveillance aircraft went from program kickoff to deployed product in under a year by using *“short schedules... and strong teamwork”*, proving that new systems *“can be developed, built, and deployed in a short period of time”* ²⁷ ²⁸. Similarly, the Marine Corps’ Harvest Hawk gunship took only 19 months from concept to combat by repurposing existing aircraft – a stark contrast to typical defense programs that span many years ²⁹. These examples underscore FIST’s core tenet: **speed is a virtue** for innovation, and it often goes hand-in-hand with simplicity and thrift ³⁰ ²⁸.

In Dinis Cruz’s Semantic Knowledge Graph work, speed and agility are achieved through automation and real-time interaction. By leveraging LLMs and serverless computing, Cruz’s approach replaces slow, static processes with **on-the-fly knowledge generation**. Rather than spending weeks designing a questionnaire or months manually gathering requirements, the AI-driven workflow can elicit and structure information from a user in a single interactive session ³¹ ²². The system adapts questions dynamically, meaning it **zeroes in on relevant details faster** than any fixed survey could ²⁴ ³². For example, if a user’s first answer indicates they operate in the healthcare sector, subsequent questions immediately drill into healthcare-specific risks, skipping irrelevant topics and saving time ³³ ³⁴. This agility mirrors the rapid iteration of Agile software development (which FIST already embraces ¹¹) but now applied to knowledge capture.

Moreover, Cruz’s use of serverless architecture means the platform can scale and respond to many users or data sources in parallel, without the delays of provisioning infrastructure. The **elastic, on-demand nature** of the system allows knowledge graphs to be built or updated whenever needed, within seconds or minutes, and then resources are freed ¹⁷ ¹⁸. This is a clear instance of “Fast” in action. In fact, tasks that previously would have been *“ridiculously expensive [and slow] to create”* due to their complexity are now *feasible, even elegant*, thanks to modern AI and graph tech ³⁵. As Cruz notes, orchestrating such adaptive dialogues across multiple data sources would have been prohibitively time-consuming just a few years ago, but today it can be done essentially *“on the fly”* ³⁵. The speed at which

personalized graphs can be generated or updated (for example, integrating new information or responding to a change in the user's context) exemplifies FIST's fast principle.

In summary, the Semantic Knowledge Graph approach achieves **rapid delivery** of knowledge structures: it turns what could be a laborious Q&A process into a swift, conversational exchange. This is very much in spirit with FIST's goal of putting capability in the user's hands quickly ³⁶. Just as FIST projects aim to *"deliver... in a short period of time"* ³⁷ ²⁸, Cruz's method delivers a usable knowledge model (e.g. a risk profile or personalized content graph) in one session rather than an extended project. Both approaches demonstrate that speed is not merely about working faster, but about **structuring the work differently** – using iterative loops, reusing existing assets, and focusing only on what's needed – to compress timelines significantly.

Cost Efficiency and Resource Constraints ("Inexpensive")

A hallmark of FIST is doing more with less. By **keeping budgets tiny**, FIST forces creativity and prioritization, often leading to innovative solutions that might be overlooked in a big-budget scenario. This principle is apparent in the earlier examples: the Condor Cluster supercomputer was built for roughly one-tenth the cost of a traditional supercomputer by cleverly using 1,760 Sony PlayStation 3 consoles running Linux (an open-source OS) instead of custom high-end hardware ²⁸. Not only did this save money, but it also slashed power consumption to one-tenth of comparable systems ³⁸ ³⁹. The Condor Cluster's success vividly illustrates that *"great things can be developed, built, and deployed... without busting the bank."* The **key was simplicity and frugality** – leveraging commodity components and open software to achieve a high-end capability ²⁸ ⁴⁰. Likewise, the Harvest Hawk program avoided the enormous expense of designing a brand-new gunship by retrofitting an existing aircraft, placing a premium on *"thrift"* (and speed) to meet the need at far lower cost ³⁰. FIST-driven teams also restrain documentation and process bloat, which indirectly saves cost by reducing man-hours spent on non-essential work ⁴¹. In short, **thriftiness** in FIST isn't about cheaping out; it's about **maximizing value for every dollar and focusing resources where they matter**.

Dinis Cruz's methodology likewise emphasizes cost efficiency, albeit in a different domain. By using **open-source and cloud-native tools**, his Semantic Knowledge Graph projects avoid the heavy licensing fees and infrastructure costs that often accompany large enterprise knowledge management solutions. The use of MGraph-DB (which stores graphs as simple JSON files) on cloud object storage means there's no need for expensive database servers – the system pays only for the storage and compute it actually uses. This serverless, pay-per-use model inherently controls costs: if no one is using the system at a given moment, it incurs virtually no expense ⁴² ¹⁸. Additionally, open standards and open-source components ensure there's **no vendor lock-in**, so organizations are not forced into pricey long-term contracts ¹⁹ ²⁰. As noted, Cruz has found that building on this stack kept operations lean while still supporting multiple knowledge graphs at once ²⁰.

Another aspect of cost efficiency is **automation replacing labor-intensive tasks**. In traditional settings, populating a knowledge base or conducting a risk assessment could involve hiring consultants or analysts for many hours. Cruz's GenAI-driven approach automates a large portion of this work: the LLM handles asking questions and parsing answers, drastically reducing the labor required to gather information. This means a small team (or even a single engineer) can deliver personalized knowledge graphs to many users simultaneously – a scenario FIST would recognize as using a *Tiny* team to achieve something that normally would demand a big support staff. Indeed, by **scaling via software and AI**, the marginal cost of serving each additional user or generating each new graph is very low. One can conduct dozens of interviews in parallel through AI, which *"used to require a team of consultants or lengthy manual research"* – now it *"can be done quickly and at scale"*, with consistency and at low

operational overhead ⁴³ ⁴⁴ . This directly speaks to the “Inexpensive” pillar: achieving affordability not just in upfront budget but in **unit cost per outcome**.

In practice, Cruz’s projects have demonstrated cost savings similar to FIST case studies (albeit on a different scale). For example, the MyFeeds.ai project (discussed later) delivered personalized content feeds without a dedicated content curation team by using automated semantic tagging and AI summarization – effectively doing the work of many analysts at a fraction of the cost. Also, by **reusing existing data and tools**, the approach channels the FIST spirit of not re-inventing the wheel. In one of Cruz’s scenarios, instead of asking a user for data that could be fetched from public sources, the system pulls it automatically (e.g. company info or threat intelligence) so that the user’s time (and thus cost) is not wasted on trivial facts ⁴⁵ ⁴⁶ . This is analogous to FIST’s reuse of existing components (like existing airframes or COTS hardware) to save cost. Overall, the Semantic Knowledge Graph methodology shows that even advanced AI-powered solutions can be built and operated **inexpensively** by adhering to lean principles: use open, existing resources, automate aggressively, and scale out only as needed.

Simplicity in Design and Complexity Management (“Simple”)

The third FIST tenet, **Simple**, is all about minimizing complexity in both the product and the process. FIST emphasizes that *complexity is not a virtue* – a complex design often signals poor understanding or over-engineering ⁹ . Simplicity, on the other hand, tends to correlate with reliability and user-friendliness. As Dan Ward put it, “*True sophistication is found in simplicity*”, whether we’re talking about system architectures or even something as mundane as PowerPoint charts ⁹ ⁴⁷ . In FIST practice, teams are encouraged to simplify requirements, organizations, and documentation. A telling example is the comparison of a 25-page fighter jet RFP versus a 26-page recipe for brownies – FIST humorously suggests that if a combat aircraft’s core spec can be kept brief, so should our other documents ⁴¹ . Simplicity also means avoiding unneeded features and gold-plating. The Harvest Hawk’s designers, for instance, made the 30mm cannon optional rather than built-in, which was “*a concrete example of engineering restraint*” that simplified the mod and logistics ³⁰ . By “*paring down requirements*” and focusing only on what truly delivers capability, FIST projects maintain clarity and avoid getting bogged down in complexity ⁴⁸ ⁴⁹ . The result is typically a solution that is easier to build, easier to use, and less prone to failure.

Cruz’s Semantic Knowledge Graph approach embodies simplicity in multiple dimensions. First, consider the **user experience**: Instead of confronting users with a massive, one-size-fits-all questionnaire or an intimidating form, the process is an intuitive conversation. Users answer one question at a time, always in context, which keeps things simple for them. They do not need to understand the underlying ontology or database structures – the AI handles that translation. In Cruz’s words, this approach turns an overwhelming interrogation into a natural dialogue, making the knowledge capture “*feel[] natural and valuable*” ⁵⁰ . This significantly reduces user fatigue and increases engagement, as evidenced by users volunteering richer information when the context is clear ³² ⁵¹ . By keeping each interaction bite-sized and contextual, the complexity of the overall task (filling in 50+ data points) is hidden behind a simple interface.

On the **design/architecture side**, simplicity is achieved through modularity and clear separation of concerns. The G³ philosophy inherently fights complexity by **breaking the problem space into many small, manageable graphs**. Each graph (for a particular domain or persona) remains relatively simple and focused. There’s no need to create a gigantic schema that covers every possible concept in the universe; instead, each context gets its own lightweight schema. This is precisely “*restraint*” applied to knowledge modeling – resisting the urge to force a single rigid structure. Cruz’s note that his system avoided forcing people to learn a standardized language is illustrative ²⁰ . It implies the solution met users where they were, using whatever terminology made sense in each case (the AI could map

terminologies between graphs). Thus, the *integration* of multiple small graphs via G³ is handled by AI “translators,” which simplifies each individual graph’s design. This stands in contrast to traditional top-down ontology projects that often become unwieldy (and stall) due to over-complexity. In effect, the Semantic Knowledge Graph approach **collapses the distinction between ontology and data** by building the ontology as it gathers data, as noted in discussions around Palantir’s methodology ⁵² ⁵³ . This collapse into a single, actionable system means there aren’t separate layers of complexity; the semantic graph is the operational representation of knowledge, kept as simple as possible to serve its purpose.

Furthermore, Cruz’s implementation choices favor simplicity: using JSON for graph storage (human-readable, easy to debug) and Python/FastAPI for orchestration (languages and frameworks known for simplicity and readability). The **simplicity of maintenance** is also evident – the system can be version-controlled and diffed like text, making changes transparent ¹⁴ . Cruz highlights the importance of being able to refactor and test graphs as key to keeping the knowledge base clean and correct over time ⁵⁴ . These engineering practices mirror how software engineers manage code complexity, bringing those benefits to knowledge engineering.

An example of simplicity from Cruz’s case studies is how **The Cyber Boardroom** project structured its security risk knowledge. Rather than enforce a massive standard taxonomy of risks and controls, the project allowed each company’s graph to organically map risks to the company’s own objectives and terminology. Yet, thanks to semantic links, all those different graphs could still be queried in a unified way (e.g. comparing two companies’ preparedness) without each having identical structure. In Cruz’s report, having each risk and control node trace up to business objectives created “*natural organizational principles without forcing rigid standardisation*”, so **each company’s graph remained unique but still interoperable** ⁵⁵ . This is a perfect example of simplicity: the structure was allowed to remain simple and relevant for each stakeholder, and complexity was avoided by not imposing unnecessary rules.

In summary, Dinis Cruz’s approach mirrors FIST’s “Simple” by prioritizing **user-friendly design, modular architecture, and avoidance of over-engineering**. Both FIST and the Semantic Knowledge Graph methodology treat simplicity not as an afterthought but as a guiding principle that influences every decision – from scoping requirements to choosing tools. And both have observed that simplicity leads to better outcomes: fewer delays, fewer misunderstandings, and systems that people actually trust and use ⁵⁶ ⁵⁷ .

Small Scale and Modular Scope (“Tiny” & Restraint)

The “Tiny” in FIST often translates to doing things on a small scale – small teams, small deliverables, tightly scoped objectives – and practicing **restraint** in growth. It recognizes that large projects can become unwieldy, so keeping efforts tiny can paradoxically achieve big results. Many FIST case studies highlight this: Instead of a 500-page specification, write a 25-page one; instead of an army of contractors, empower a small skilled team. The FIST comics famously personified this value as *Lt. Nano*, a 6-inch-tall superhero proving that even tiny forces can defeat giant problems ⁵⁸ ⁵⁹ . In real terms, “Tiny” means avoiding bloated project scopes and feature creep. Requirements are kept to the minimal viable set needed for the mission, and any additional “nice-to-haves” are deferred or discarded. FIST advocates for *fixed schedules with floating requirements* as a form of restraint ⁵ – essentially capping how long and how large a project can get, and adjusting scope to fit those caps. This disciplined smallness is often the hardest cultural shift, because it runs counter to the desire for comprehensive solutions. Yet, as FIST shows, a series of small, focused successes is better than one grandiose failure. Tiny also implies **modularity**: it’s easier to deliver something tiny that works and then build on it, than to attempt a gigantic deliverable in one go.

Cruz's G³ concept is inherently about "tiny" in the sense of **modular knowledge graphs**. Each graph in the G³ network can be seen as a "tiny" slice of the overall knowledge needed. For example, one graph might represent a CISO's perspective on cybersecurity risks, another graph might represent a CTO's view of technology assets, and another captures regulatory requirements. Instead of merging these into one giant schema (which would grow complex and unwieldy), G³ links them at a higher level. This exhibits restraint: each graph is allowed to stay small and relevant to its owner, and only the necessary connections are established. It is essentially the microservices philosophy applied to knowledge management – many small services (graphs) that together fulfill larger goals. As a result, adding or updating one piece doesn't require overhauling the whole, and failures or changes in one domain don't cascade uncontrollably to others. This aligns perfectly with FIST's notion that *smaller is more manageable and often more effective*. In Cruz's workflow, when building a persona's knowledge graph, the process literally **starts tiny** (one node) and grows only as needed with each answer ²² ²⁴ . If a piece of data is not relevant, it never enters the graph – keeping the graph's scope as small as it can be while still being useful. This incremental growth is a form of disciplined restraint akin to FIST's fixed schedule: the interview might be time-boxed or limited in number of questions, so the system must decide the *most important* things to include and skip the rest ⁶⁰ ⁶¹ . The outcome is a *"rich graph that truly captures important nuances, rather than a shallow one-size-fits-all dataset."* ⁶² ⁶³ . In other words, each graph is only as big as it needs to be.

We also see "Tiny" in how Cruz's solutions are implemented organizationally. These projects do not require large teams to manage; a handful of developers and domain experts can deploy the system for many users. The knowledge graphs produced are often concise representations (e.g. a few hundred nodes to represent an organization's risk profile, which is tiny compared to sprawling relational databases of old). And because each user or stakeholder only deals with their portion of the graph, the cognitive load stays small. This resonates with FIST's emphasis on small, empowered teams: Cruz's method allows **decentralizing knowledge capture** (each user contributes to their graph) rather than funneling everyone into one massive data call.

A concrete parallel: Consider the **Harvest Hawk** example vs. Cruz's approach to content feeds. Harvest Hawk avoided building a huge new aircraft program (tiny scope) by adding a minimal kit to an existing plane – effectively a *modular add-on*. In Cruz's **MyFeeds.ai** personalized news system, instead of building a one-size-fits-all news algorithm, they created separate persona profiles (graphs) like "Tech CEO" or "Security Analyst," each with its own taxonomy of interests ⁶⁴ . The system then fed relevant articles into each micro-graph and generated summaries, tailoring output to that persona ⁶⁴ ⁶⁵ . Each persona graph is tiny and focused (e.g. one might heavily feature technology topics, another compliance topics), and the *flows* that manage them are modular. This modular design made it easy to add a new persona – just start a new small graph – without overhauling the entire system. It's a clear case of **restraint by design**: not trying to have one monolithic model of "what every reader cares about," but acknowledging individual differences via small modules.

In summary, the "Tiny" element of FIST – the disciplined limitation of scope and size – finds strong resonance in Dinis Cruz's Semantic Knowledge Graph and G³ work. Both approaches prove that focusing on the *smallest effective solution* yields better outcomes. By keeping knowledge representations small and modular, Cruz's approach ensures flexibility and manageability, very much in the same way FIST kept projects lean to ensure they stayed on track. Both demonstrate that *restraint is not a weakness but a strategic advantage*: it prevents waste, encourages innovation (within constraints), and allows continuous improvement on a solid, simple base.

Concrete Examples: FIST and Semantic Graphs in Action

To cement the comparison, here we highlight a few concrete examples side-by-side, illustrating how FIST principles and Dinis Cruz's methods both lead to innovative outcomes:

- **Military “Instant Gunship” vs. AI-Driven Risk Graph:** The Marine Corps' *Harvest Hawk* project converted a KC-130 tanker into a gunship in 19 months by reusing existing hardware and keeping the add-ons minimal. It exemplified FIST by placing “*a premium on simplicity, thrift, speed, and restraint*” – using a fraction of the cost and time of a typical program ³⁰. Analogously, in the *Cyber Boardroom* project, Cruz built an executive risk knowledge graph through a short interactive dialogue instead of a months-long consulting project. The resulting graph gave a comprehensive view of a company's risks and controls, tailored to the CEO's needs. This was achieved with a small team and open tools, demonstrating that even high-level strategic insight can be delivered fast and inexpensively when the approach is simplified and focused. Notably, each risk node in the graph was linked to business objectives, which created a clear, *simple model of how security relates to value*, but **without** enforcing a large standardized framework on all companies ⁵⁵. Each company's risk graph remained *tiny and specific* to them, yet thanks to semantic connections, the insights were actionable and comparable – much like *Harvest Hawk* was a custom solution for a specific need, but built from general components.
- **Condor Cluster Supercomputer vs. MyFeeds Personalized Knowledge:** The Air Force's *Condor Cluster* combined 1,760 PlayStation 3 consoles to create a record-breaking supercomputer at about 10% of the usual cost and power ³⁸ ²⁸. This achievement was a triumph of frugality and smart reuse of simple components (gaming consoles) to outperform far more expensive machines. In parallel, Cruz's *MyFeeds.ai* project tackled a complex problem – personalized information overload – with a simple, frugal approach: it uses semantic graphs plus GenAI to deliver custom news feeds. Instead of expensive proprietary recommender systems, *MyFeeds* relies on open-source graph algorithms and LLM summarizers. Each user persona is represented as a small graph of interests, and news articles are ingested into a knowledge graph of topics. A lightweight AI “agent” then matches articles to personas, even generating brief summaries for convenience ⁶⁴ ⁶⁵. The entire pipeline leverages existing tech (open APIs, NLP models) and can run on commodity cloud infrastructure. In essence, like *Condor Cluster*'s use of PS3s, *MyFeeds* repurposed readily available AI components to achieve a result (personalized, up-to-date news briefings) that would traditionally require significant investment. And it did so with *tiny modules*: separate graphs per interest profile, rather than one mega-system attempting to cater to all. This modular design kept complexity low and made the solution **fast to implement and modify** – new interest domains could be added by simply plugging in another small graph.
- **Agile Mission Software vs. Semantic DevOps:** A broader example is how FIST and Cruz's methodology both integrate with agile, iterative workflows. FIST, as noted, explicitly draws on Agile and Lean techniques to keep development cycles short and improvements continuous ¹¹. We see this in how FIST programs often delivered a minimum viable capability first, then iterated (for example, adding an optional cannon to *Harvest Hawk* later rather than delaying initial deployment ³⁰). In Cruz's world, the knowledge graphs produced are **living artifacts** – never truly “finished” but continually refined. Users can come back to update their persona graph as situations change, and the system itself can pull in new data and ask follow-up questions at intervals ⁶⁶ ⁶⁷. This is essentially *Agile for knowledge*: deliver a working knowledge graph quickly, then update it frequently in small increments. For instance, if a new cybersecurity threat emerges, the system can prompt the user about it and adjust the graph rather than redoing a whole assessment from scratch ⁶⁶ ⁶⁸. This incremental maintenance mirrors how Agile software projects release new features in sprints – it keeps the knowledge continuously relevant

(fast updates) and avoids the large, infrequent overhauls (which often fail due to complexity). FIST's influence is clear here: by treating knowledge capture as an iterative, restrained process (never letting it balloon unmanageably), Cruz's approach ensures the end product stays useful and truthful over time.

These examples reinforce the parallel: whether it's building a physical system or a knowledge system, adhering to **fast, inexpensive, simple, tiny** leads to solutions that are efficient and effective. Harvest Hawk and Condor Cluster showed that small, resource-conscious teams can beat the odds in defense projects. In the same spirit, the Semantic Knowledge Graph applications show that small-scale, adaptive approaches can solve problems in AI and knowledge management that traditional big projects have struggled with. Both domains benefit from **clever reuse, modularity**, and a focus on the core mission rather than extraneous bells and whistles.

Actionable Takeaways

Both FIST and Dinis Cruz's Semantic Knowledge Graph methodology provide lessons that can guide successful projects beyond their original contexts. Here are some actionable takeaways for practitioners seeking to emulate these principles:

- **Start Small and Iterate:** Define the minimal viable product or knowledge model that delivers value, and build that first. Whether it's a first combat-capable prototype or a basic semantic graph, getting something working quickly is better than aiming for perfection years later ³⁷₅₀. Use feedback from real use to guide the next iteration.
- **Leverage Existing Resources:** Before reinventing anything, ask what existing assets can be repurposed. It could be commercial hardware (as in Condor Cluster using PS3s) or open-source software and AI models (as in Cruz's use of existing LLMs and graph tools) ²⁸₂₅. Reusing proven components not only saves cost but also reduces integration risk.
- **Keep it Simple (Design for Clarity):** Embrace the mantra *"simple is sophisticated"*. Trim unnecessary features and complexity in design. Enforce brevity in specifications and schemas – e.g., don't force a huge ontology if a lightweight, evolving schema works ⁹₁₄. A clear, simple architecture will be easier to build, maintain, and adapt. If a process or model seems overly complex, step back and see if it can be broken down or simplified (perhaps by splitting into smaller pieces or using simpler language).
- **Enforce Constraints Deliberately:** Set hard limits on timelines, budgets, or scope at the outset – these constraints drive creativity and prioritization. FIST showed that fixed schedules with floating requirements yield faster delivery ⁵. In knowledge projects, one might constrain the number of questions or the time for an AI interview, forcing the system to focus on what's most important. Resist scope creep by regularly asking, *"Do we really need this now?"* and deferring nice-to-haves.
- **Empower Modular Teams/Components:** Structure work such that small teams or components can progress in parallel. In practice, this could mean different feature crews in a software team, or different semantic graphs for different domains, all loosely coupled. Ensure there are well-defined interfaces (in tech, APIs or data formats; in teams, communication protocols) so that these modules can later connect (G³ style) without a centralized bottleneck ¹⁵₂₁. This modular approach keeps each unit tiny and manageable, and the overall system flexible.

- **Integrate User Feedback Continuously:** Keep the end-user in the loop to validate and refine the output. FIST projects valued end-user focus – delivering capability to the field quickly so users could benefit and give feedback ³⁶. Likewise, Cruz’s approach literally involves the user at each step (answering questions, confirming data) ⁵⁶ ⁵⁷. This not only builds trust but ensures the solution remains relevant and correct. Design your process to incorporate user testing, feedback cycles, or human-in-the-loop checks rather than treating it as a one-shot deliverable.
- **Foster a Culture of “Fast and Frugal” Innovation:** Perhaps most importantly, encourage the mindset that doing things quickly and cheaply is not about cutting corners, but about *finding smarter ways*. Celebrate solutions that achieve objectives with less – less time, less money, less complexity – as successes. When team members propose elaborate, expensive plans, challenge them with FIST-like questions: *Can we do this faster? Simpler? With fewer resources?* Often, this will spur creative problem-solving and lead to alternative approaches (for example, using AI to automate a task instead of hiring more people, or simplifying a feature set to hit an early deadline).

By applying these takeaways, organizations can hope to replicate the kind of success seen in both the FIST case studies and Dinis Cruz’s projects: high-impact results delivered efficiently. In essence, it’s about **combining the discipline of constraints with the agility of modern technology** to unlock innovation.

Conclusion

The comparison between the FIST framework and Dinis Cruz’s Semantic Knowledge Graph (SKG) & G³ methodology reveals a deep commonality: both advocate for smarter, leaner ways to achieve complex goals. What started as a mantra for defense acquisitions – *Fast, Inexpensive, Simple, Tiny* – finds a natural echo in the realm of semantic software engineering. By aligning theory with practice, we see that FIST’s core values are not antiquated ideals but living principles evident in cutting-edge projects today.

In theory, FIST provides a mindset to **balance needs with resources swiftly** and effectively ³⁶. In practice, Cruz’s work shows that embracing those same values (speed, thrift, simplicity, restraint) can transform how we build and use knowledge systems. The Semantic Knowledge Graph approach, supercharged by GenAI, essentially fulfills a FIST-like manifesto in the knowledge domain: it delivers timely insights (Fast) at low cost (Inexpensive) using straightforward, adaptive models (Simple) and granular modular data (Tiny). Each conversation-driven graph is a proof that even as data and AI systems grow, we do not have to succumb to complexity or bloated design.

Furthermore, this alignment is **mutually illuminating**. FIST’s success stories like Harvest Hawk or the Condor Cluster exemplify the power of constraints and focused design – lessons that can guide AI and software projects to avoid waste and over-engineering. Conversely, the success of Cruz’s G³ approach in rapidly evolving fields like cybersecurity hints at how FIST principles can be extended with modern tools: automation and AI can enforce speed and simplicity in ways not previously possible, making it even easier to adhere to those principles at scale ⁴⁴ ⁶⁹.

A conceptual mapping between the two shows that each FIST component has its analogue in SKG/G³: **“Fast” is mirrored by real-time LLM-driven development and quick iterations, “Inexpensive” by open-source utilization and serverless efficiency, “Simple” by human-centric design and ontology-light graphs, and “Tiny” by modular graph units and restrained scopes**. These mappings aren’t just theoretical – we’ve seen them in concrete outcomes ranging from faster threat modeling sessions to cost-effective personalized content delivery, all aligning with the FIST ethos.

In closing, the work of Dinis Cruz and others on Semantic Knowledge Graphs with G³ can be viewed as a contemporary reimagining of FIST for the age of AI and big data. It demonstrates that even as our challenges evolve – be it making sense of vast information or building the next-generation systems – the solution often lies in **going small, staying simple, moving fast, and spending smart**. By fusing user-driven inputs with machine-driven scalability, we are, in effect, keeping the human ingenuity and prudence at the center (as FIST intended) while leveraging technology to amplify our ability to deliver quickly and efficiently. As one commenter insightfully noted about Cruz’s research, the shift to dynamic, self-evolving graphs is “*a compelling opportunity not only to map knowledge but also to enhance real-time decision-making*”⁵⁷ – exactly the kind of high payoff that FIST would predict from a fast, simple approach. Both FIST and the SKG/G³ methodology remind us that **innovation flourishes under constraints** when guided by clear principles. By learning from both, future projects in any field – whether building airplanes or knowledge graphs – can aim to be *faster, cheaper, simpler, and smaller...* and in so doing, often achieve better outcomes for all.

Sources: The FIST framework as described by Lt. Col. Dan Ward (Defense AT&L, 2006–2011)^{5 28}; examples from Defense programs (Harvest Hawk, Project Liberty, Condor Cluster)^{30 28}; Dinis Cruz’s published research on Semantic Knowledge Graphs and G³ (2023–2025)^{15 20}; and content from *User-Driven Semantic Persona Graphs Powered by GenAI*, co-authored by Cruz and ChatGPT Deep Research^{16 44}. These sources collectively illustrate and support the analysis above, showing a convergence of ideas from defense acquisitions and modern AI knowledge engineering.

¹ Dan Ward's Corner - AcqNotes

<https://acqnotes.com/danward>

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¹⁵ This is exactly why my focus is on G3 (Graphs of Graphs of Graphs), where we have ontologies of Ontologies and Taxonomies of Taxonomies | Dinis Cruz
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